if condition

In [1]:
```
if True:  # indentiation is always 4 spaces
    print('Data Science')
```

```
Data Science
```

In [2]:
```
if True:
    print('Data Science')
    print('Hi Shreya')
```

```
Data Science
Hi Shreya
```

In [3]:
```
if False:
    print('Data Science')
print('good bye')
```

```
good bye
```

In [4]:
```
if True:
    print('Data Science')
print('good bye')
```

```
Data Science
good bye
```

if else

In [5]:
```
if True: # indentiation is always 4 spaces
    print('Data Science')
else:
    print('good bye')
```

```
Data Science
```

In [6]:
```
if False: # indentiation is always 4 spaces
    print('Data Science')
else:
    print('good bye')
```

```
good bye
```

In [20]:
```
x = 5
r = x % 2

if r == 0:
    print('Even number')
else:
    print('odd number')
```

```
odd number
```

In [ ]:

print only even number

In [7]:
```
x = 4
r = x % 2

if r == 0:
    print('Even number')
```

```
Even number
```

In [8]:
```
x = 5
r = x % 2

if r == 0:
    print('Even number')
```

In [9]:
```
x = 9
r = x % 2

if r == 0:
    print('Even number')

if r == 1:
    print('odd number')
```

```
odd number
```

In [10]:
```
x = 8
r = x % 2

if r == 0:
    print('Even number')
else:
    print('odd number')
```

```
Even number
```

In [11]:
```
x = 5
r = x % 2

if r == 0:
    print('Even number')
print('odd number')
```

```
odd number
```

In [12]:
```
x = 4
r = x % 2

if r == 0:
    print('Even number')
print('odd number')
```

```
Even number
odd number
```

In [13]:
```
x = 4
r = x % 2
```

```
if r == 0:
    print('Even number')
else:
    print('odd number')
```

```
Even number
```

multi if

In [14]:
```
x = 17
r = x % 2

if r == 0:
    print('Even number')

if r != 0:
    print('odd number')
```

```
odd number
```

In [21]:
```
x = 4
r = x % 2

if r == 0:
    print('Even number')

if r == 1:
    print('odd number')

if r == 2:
    print('even number')
```

```
Even number
```

In [22]:
```
x = 1

if x == 1:
    print('one')
if x == 2:
    print('Two')
if x == 3:
    print('Three')
if x == 4:
    print('four')
```

```
one
```

nested if

In [15]:
```
x = 3
r = x % 2

if r == 0:
    print('Even number')
    if x>5:
        print('greater number')
```

```
else:
    print('Odd Number')
```

```
Odd Number
```

In [16]:
```
x = 4
r = x % 2

if r == 0:
    print('Even number')
    if x>5:
        print('greater number')

else:
    print('Odd Number')
```

```
Even number
```

In [17]:
```
x = 6
r = x % 2

if r == 0:
    print('Even number')
    if x>5:
        print('greater number')

else:
    print('Odd Number')
```

```
Even number
greater number
```

In [23]:
```
x = 4
r = x % 2

if r == 0:
    print('Even number')

    if x>5:
        print('greater number')


    else:
        print('number is lesser number')

else:
    print('Odd Number')
```

```
Even number
number is lesser number
```

In [18]:
```
x = 2
r = x % 2

if r == 0:
    print('Even number')

    if x>5:
        print('greater number')
    else:
        print('not greater')
```

```
else:
    print('Odd Number')
```

```
Even number
not greater
```

In [19]:
```
year = int(input("Enter a year: "))

if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
 print("Leap Year")
else:
 print("Not a Leap Year")
```

```
Not a Leap Year
```

if elif

In [24]:
```
# elif it wont check till the block once you find the output it wont go to next
# you can try with multiple parameter 1, 2 & 3 value in x

x = 4

if(x == 1):
    print('one')
elif(x == 2):
    print('Two')
elif(x == 3):
    print('Three')
elif(x == 4):
    print('four')
```

```
four
```

In [25]:
```
# elif it wont check till the block once you find the output it wont go to next
# you can try with multiple parameter 1, 2 & 3 value in x

x = 7

if(x == 1):
    print('one')
elif(x == 2):
    print('Two')
elif(x == 3):
    print('Three')
elif(x == 4):
    print('four')
```

In [26]:
```
# elif it wont check till the block once you find the output it wont go to next
# you can try with multiple parameter 1, 2 & 3 value in x

x = 7

if(x == 1):
    print('one')

elif(x == 2):
    print('Two')
elif(x == 3):
    print('Three')
```

```
elif(x == 4):
    print('four')

else:
    print('number not found')
```

```
number not found
```

In [ ]: